

# EE 102 Week 3, Lecture 2 (Fall 2026)

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# 1 Goals for today

- Review (and wrap up) the introduction to signals and systems.
- Appreciate the importance of the complex exponential signal in signal processing and communications.
- Understand the properties of systems: time invariance, linearity, causality, memory, invertibility.
- Analyze system response using linearity and time-invariance.

## 2 Review: Signals and Systems

Recall that signals are mathematical functions that represent physical quantities, or information. Usually, we may think of signals as functions of time, but they can also be functions of space or other variables. A simple rule of thumb to remember is that signals are mathematical functions (that you can also visualize, if they are 1D or 2D).

On the other hand, you can easily distinguish systems from signals because systems are “processors” of signals. Every quantity/information/measurement is a signal, and systems can process a signal to produce an output signal. The signal that gets processed by the system is the input signal and the processed signal is the output signal. Due to this, a common abstraction of a system is a “black box” that takes in an input signal and produces an output signal. Formally, we write a system as an operator (or a mapping) that takes in a signal and produces another signal:

$$H : x(t) \mapsto y(t)$$

where  $x(t)$  is the input signal,  $y(t)$  is the output signal, and  $H$  is the system. Note that the above does not imply that  $x(t) = y(t)$ , and in fact, if the system is doing something meaningful, the output signal will be different from the input signal. System examples:

### 2.1 Example: A matrix as a system

Given an input vector  $x \in \mathbb{R}^n$ , a matrix  $A \in \mathbb{R}^{m \times n}$  can be thought of as a system that produces an output vector  $y \in \mathbb{R}^m$  as follows:

$$y = Ax$$

where  $A$  is the system that performs the matrix multiplication operation. The block diagram of the system can be represented as:

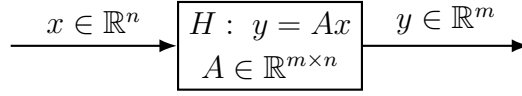


Figure 1: A system as a linear map implemented by matrix multiplication.

## 2.2 Example: An amplifier as a system

An amplifier is a system that takes in an input signal  $x(t)$  and produces an output signal  $y(t)$  by amplifying the input signal by a constant factor  $\alpha \in \mathbb{R}$ . The system can be represented as:

$$H : x(t) \mapsto y(t) = \alpha x(t)$$

where  $H$  is the system that amplifies the input signal by the factor  $\alpha$ . We can draw a block diagram of the system as follows:

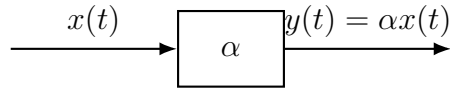


Figure 2: Amplifier system: multiply the input by a real gain  $\alpha$ .

## 2.3 Example: A differential equation as a system

A system can also be represented by a differential equation that relates the input signal  $x(t)$  and the output signal  $y(t)$ . For example, consider the following first-order linear differential equation:

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

where  $a$  and  $b$  are constants. This equation describes a system that takes in the input signal  $x(t)$  and produces the output signal  $y(t)$  by solving the differential equation. In this case, the block diagram of the system can be represented as:

## 2.4 Example: A frequency modulation system

A frequency modulation (FM) system is a system that takes in an input signal  $x(t)$  and produces an output signal  $y(t)$  by modulating the frequency of a carrier signal.

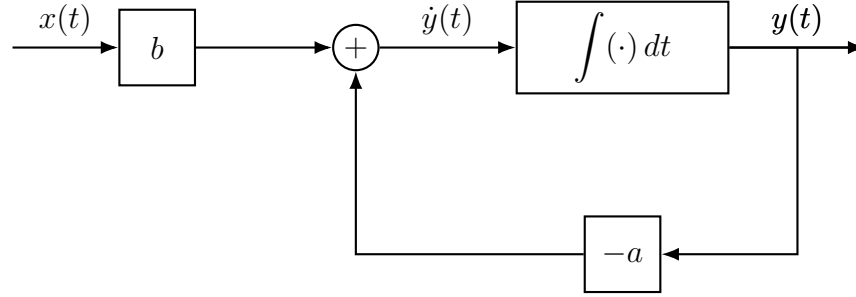


Figure 3: Realization of  $\dot{y}(t) + a y(t) = b x(t)$ .

### Pop Quiz 2.1: Check your understanding!

For a purely sinusoidal audio signal  $x(t) = A \cos(\omega_0 t)$ , define a frequency modulator system that produces an output signal  $y(t)$  that has a different frequency than the input signal. What are various ways in which you can achieve this? Can you think of a general approach to create such a system by leveraging the properties of the complex exponential signal (see previous chapter of the notes)?

*Solution on page 15*

Consider a specific example where  $x(t) = e^{j2\pi(10t)}$  and we send this input through a modulator system that multiplies  $e^{j2\pi(32t)}$  to the input signal. The output can be obtained by computing the shift in frequency (exponents get added on multiplication). The output signal  $y(t)$  for both the real and imaginary parts of the complex exponential modulator system is shown in Figure 4.

An audio signal can be thought of as a combination of many sinusoidal signals with different frequencies. So, we can leverage the properties of the general complex exponential signal to define the input audio signal  $x(t)$  as a sum of complex exponentials:

$$x(t) = \sum_k \operatorname{Re} (A_k e^{j\omega_k t})$$

where  $A_k$  and  $\omega_k$  are the amplitude and frequency of the  $k$ -th sinusoidal component of the audio signal. Note that for simplicity, we are just considering the cosine component. To keep it general, we can continue to work with complex numbers as it allows us to take into account both sine and cosine components (orthogonally phase shifted by  $\pi/2$ , which is important in representing a general signal). If we define our system as a multiplication of the input signal with  $\operatorname{Re}\{Ae^{j\omega_0 t}\}$ , we will see that the output signal has a frequency that is shifted by  $\omega_0$  from the input signal. The system can be represented as:

$$H : x(t) \mapsto y(t) = x(t) \cdot \operatorname{Re}\{Ae^{j\omega_0 t}\}$$

where  $H$  is the system that modulates the frequency of the input signal by multiplying it with a carrier signal  $\operatorname{Re}\{Ae^{j\omega_0 t}\}$ . In this case, the output  $y(t)$  can be written as a sum of

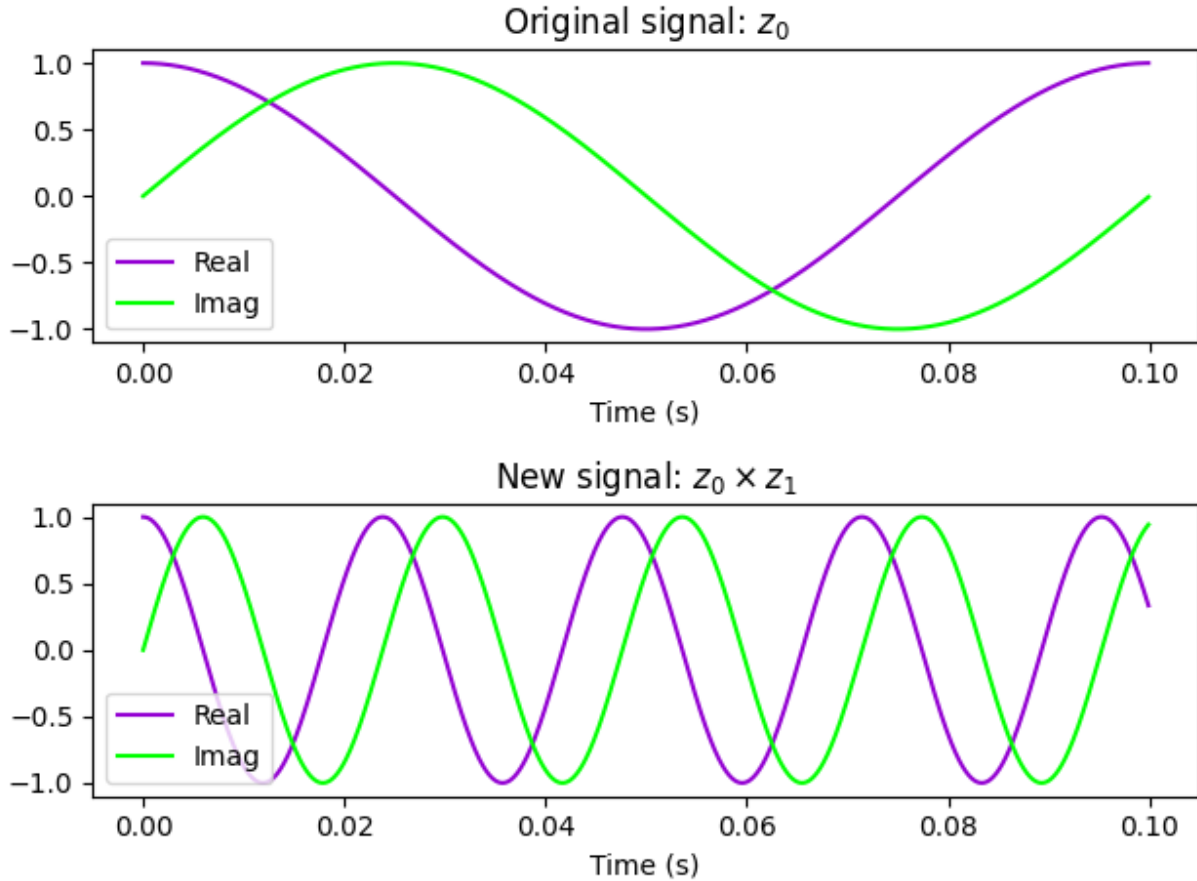


Figure 4: Frequency modulation of a sinusoidal signal using multiplication with a complex exponential.

sinusoidal signals with frequencies shifted by  $\omega_0$ :

$$y(t) = \sum_k \text{Re} (A_k e^{j(\omega_k + \omega_0)t})$$

For a real audio signal that you can find on course Github: [guitar\\_clean.wav](#), we can create the simple frequency modulator using the multiplication of a complex exponential. The input and the output signals are shown in Figure 5. You can interactively visualize and interpret the frequency of the output signal by running the virtual manipulator on frequency modulator using complex exponentials: [VM\\_modulation.py](#).

### Pop Quiz 2.2: Check your understanding!

By using the virtual manipulator [VM\\_modulation.py](#), test three different frequencies of the modulator system: 0.5 Hz, 5 Hz, and 20 Hz. Reflect on the expected sound for

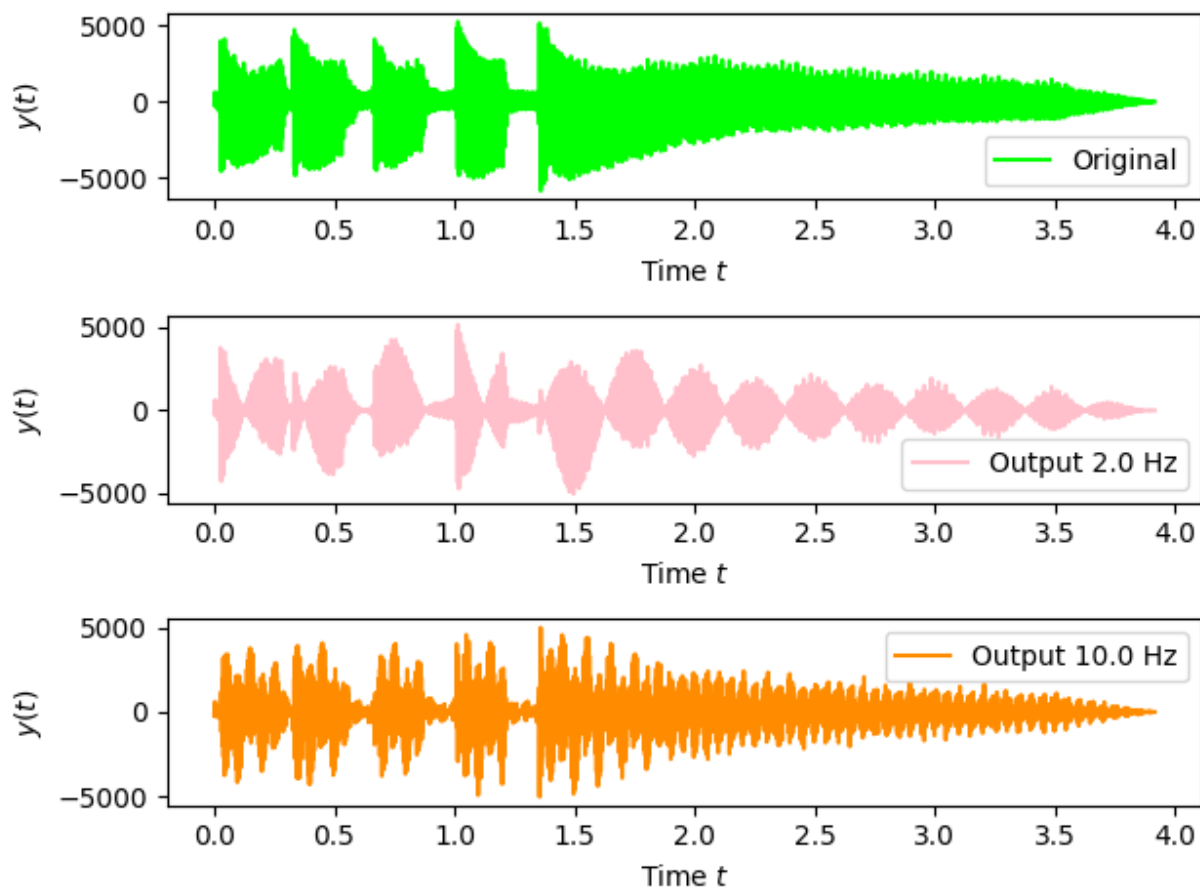


Figure 5: Frequency modulation of a guitar audio signal using multiplication with a complex exponential.

each modulation before listening to the output `guitar_modulated.wav`. Which one of the output signals is closest to the original audio signal? How would you describe the frequency modulation effect on audio?

*Solution on page 15*

### Pop Quiz 2.3: Check your understanding!

Can you define a noise cancellation system that takes in a noisy signal and produces an output signal that cancels the noise? Assume that the noise is a sinusoidal signal with a known frequency and amplitude.

*Solution on page 15*

### 3 Review: The three special signals

We noted three special signals: step, impulse, complex exponential and gave them a name (the “timeless trio”). We will now review these three signals in more detail.

#### 3.1 The unit step as a switch

The unit step  $u(t)$  is a fundamental building block for constructing piecewise signals. It acts as a switch that turns on at  $t = 0$  and remains on thereafter. By combining and shifting unit steps, we can create complex piecewise functions.

The ideal unit step function is not realizable in practice. However, smooth *sigmoids* can approximate  $u(t)$ . You will find that these sigmoids are the main building blocks of neural networks! In a reductionist view, neural networks are just millions (or billions) of such sigmoids (unit steps) trained together to approximate any general function. That’s why neural networks are sometimes also called *universal function approximators*. Someone who learns signal processing will be able to see the deeper picture: everything is a switch after all (as in electronics and computers).

You can write down the sigmoid functions as approximate unit steps:

$$\sigma_k(t) = \frac{1}{1 + e^{-kt}} \quad \text{and} \quad \tilde{\sigma}_k(t) = \frac{1}{2}(1 + \tanh(kt)), \quad k \gg 1 \Rightarrow \sigma_k(t) \approx u(t).$$

Let’s consider two examples to demonstrate the power of the unit step functions.

#### 3.2 Example: A square wave using steps

The bipolar square pulse in Figure 6 can be written as a combination of unit step functions. Let’s try to build it step-by-step. First, note that the first unit step (part 1 of the figure) turns on the signal at  $t = 0$ , so this can be written as  $u(t)$ . Next, we want the signal to turn off at  $t = 1$  (part 2 of the figure). Consider the time-shifted signal  $u(t - 1)$ , which turns on at  $t = 1$ . If we subtract this from the first step, we get a signal that is +1 on  $[0, T)$  and 0 elsewhere. Next, we want the signal to turn negative at  $t = 2T$  (part 3 of the figure). Consider the negative of the step function  $-u(t - 2T)$ , which turns on at  $t = 2T$ . Adding this to the previous result gives a signal that is +1 on  $[0, T)$ , 0 on  $[T, 2T)$ , and -1 on  $[2T, \infty)$ . Finally, we want the signal to return to 0 at  $t = 3T$  (part 4 of the figure). Consider the time-shifted step  $u(t - 3T)$ , which turns on at  $t = 3T$ . Adding this to the previous result

gives the desired square pulse. Thus, the complete expression for the square pulse is

$$x(t) = u(t) - u(t - T) - u(t - 2T) + u(t - 3T)$$

equals  $+1$  on  $[0, T)$ ,  $0$  on  $[T, 2T)$ ,  $-1$  on  $[2T, 3T)$ , and  $0$  otherwise. It is a sum of four shifted/scaled steps. Figure 6 overlays the four contributors.

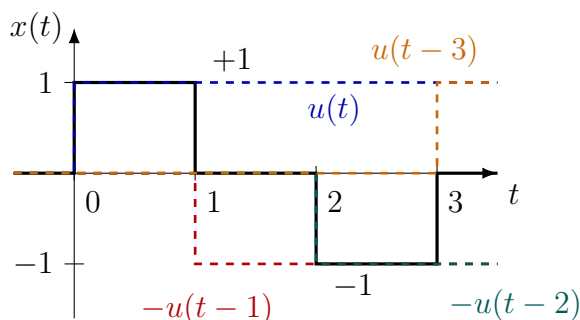


Figure 6: Square pulse constructed as  $u(t) - u(t - T) - u(t - 2T) + u(t - 3T)$  (here  $T = 1$ ). Colored dashed curves show the four step contributors.

**Example 2 (piecewise levels using steps).** A signal that is 0 for  $t < 1$ , jumps to 1 on  $[1, 2)$ , and then to 2 for  $t \geq 2$  can be written *only* with steps:

$$x(t) = u(t - 1) + u(t - 2).$$

More generally, any piecewise-constant  $x(t)$  can be expressed as  $\sum_k a_k u(t - t_k)$  with appropriate increments  $a_k$  at the change points  $t_k$ .

### 3.3 The unit impulse as a sampler

We defined the impulse by its sifting property:

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0).$$

Think of  $\delta$  as an idealized sampler that extracts the value of  $f(t)$  at  $t_0$ .

A common approximation uses a narrow rectangle (area 1):

$$p_\varepsilon(t) = \frac{1}{\varepsilon} \left[ u\left(t + \frac{\varepsilon}{2}\right) - u\left(t - \frac{\varepsilon}{2}\right) \right], \quad \lim_{\varepsilon \rightarrow 0^+} p_\varepsilon(t) = \delta(t) \text{ in the distribution sense.}$$

You can visualize this approximation by running the [VM\\_delta\\_via\\_pulse.py](#) script available on course Github.

## 4 A candidate for “most” special in signal processing: Complex exponentials

You have seen the complex exponential function before:  $e^{j\omega t}$ . However, that is just a special case of the generalized complex exponential

$$x(t) = Ae^{st}, \quad A \in \mathbb{C}, \quad s = \sigma + j\omega \in \mathbb{C}.$$

### 4.1 Why complex numbers / complex signals?

With complex numbers, the central question is always: why did we even introduce the imaginary! Recall that when we worked with 2D vectors, we could represent them instead using a complex scalar. So, the vector operations, which can be tricky to account for become simpler with the standard scalar algebra that is applicable for complex numbers. We want the same convenience with signals. Additionally, as we will see, complex exponentials let us represent a *really* large class of signals such as sinusoids (of all kinds!), exponentials (both growing and decaying), and arbitrary combinations of these!

#### Pop Quiz 4.1: Check your understanding!

Recall that the utility of complex numbers is that they allow us to represent higher dimensions, in this case, two signals, using just one signal. Can you identify the two signals represented by  $x(t) = e^{j\omega t}$ ?

*Solution on page 15*

To see that the general complex exponential can represent a large class of signals, write  $A = |A|e^{j\phi}$  to obtain the *amplitude-phase* form

$$x(t) = |A|e^{\sigma t}e^{j(\omega t + \phi)} = e^{\sigma t}(|A|\cos(\omega t + \phi) + j|A|\sin(\omega t + \phi)).$$

Special cases emerge by selecting  $A$  and  $s$ :

### 4.2 Example 1: A sinusoidal signal using complex exponentials

Set  $\sigma = 0$ ,  $A = |A|e^{j\phi}$ ,  $s = j\omega$ :

$$x(t) = |A|e^{j(\omega t + \phi)} \Rightarrow \operatorname{Re}\{x(t)\} = |A|\cos(\omega t + \phi), \quad \operatorname{Im}\{x(t)\} = |A|\sin(\omega t + \phi).$$

You can see that  $e^{j\omega t}$  is periodic with  $T = \frac{2\pi}{\omega}$  for  $\omega \neq 0$ .

### Pop Quiz 4.2: Check your understanding!

When is  $Ae^{st}$  a constant  $k$ ?

*Solution on page 15*

## 4.3 Example 2: Real decaying exponential

Take  $\omega = 0$ ,  $\sigma < 0$ , and  $A \in \mathbb{R}$ :

$$x(t) = Ae^{\sigma t}, \quad \text{monotone decay to 0 as } t \rightarrow \infty.$$

### Pop Quiz 4.3: Check your understanding!

Derive the growing exponential using the complex exponential.

*Solution on page 15*

## 4.4 Example 3: Exponentially damped sinusoid

Choose  $\sigma < 0$ ,  $\omega \neq 0$ ,  $A = |A|e^{j\phi}$  and take the real part:

$$x(t) = |A|e^{\sigma t} \cos(\omega t + \phi).$$

### Pop Quiz 4.4: Check your understanding!

Derive the exponentially growing cosine using the complex exponential.

*Solution on page 15*

## 4.5 A reminder about complex algebra

You can derive all of the standard signals above from the general complex exponential also by expanding out the complex exponential by writing  $A = a_1 + jb_1$  and  $s = \sigma + j\omega$ :

$$x(t) = (a_1 + jb_1)e^{(\sigma + j\omega)t} = (a_1 + jb_1)e^{\sigma t}(\cos(\omega t) + j\sin(\omega t)).$$

Expanding this out, we get

$$x(t) = e^{\sigma t}((a_1 \cos(\omega t) - b_1 \sin(\omega t)) + j(a_1 \sin(\omega t) + b_1 \cos(\omega t))).$$

Thus, the real and imaginary parts of  $x(t)$  are

$$\operatorname{Re}\{x(t)\} = e^{\sigma t}(a_1 \cos(\omega t) - b_1 \sin(\omega t)), \quad \operatorname{Im}\{x(t)\} = e^{\sigma t}(a_1 \sin(\omega t) + b_1 \cos(\omega t)).$$

This shows that by choosing appropriate values for  $a_1$ ,  $b_1$ ,  $\sigma$ , and  $\omega$ , we can generate a wide variety of signals, including sinusoids, exponentials, and combinations thereof.

## 5 Properties of Systems

Systems can have various properties that define their behavior and characteristics. Some of the most common properties of systems are:

- **Linearity:** A system is linear if it satisfies the principles of superposition and homogeneity. This means that the output of the system for a linear combination of input signals is equal to the same linear combination of the outputs for each individual input signal.
- **Time-invariance:** A system is time-invariant if its behavior and characteristics do not change over time. This means that if the input signal is shifted in time, the output signal will also be shifted by the same amount.
- **Causality:** A system is causal if the output at any given time depends only on the current and past input values, and not on future input values. In other words, a causal system cannot anticipate future inputs.
- **Memory:** A system has memory if its output depends on past input values. A system without memory (also called memoryless) produces an output that depends only on the current input value.
- **Invertibility:** A system is invertible if there exists an inverse system that can recover the original input signal from the output signal. In other words, if you apply the inverse system to the output, you should get back the original input.
- **Stability:** A system is “BIBO” stable if bounded input signals produce bounded output signals. This means that if the input signal remains within a certain range, the output signal will also remain within a certain range (bounded input implies bounded output: BIBO).

### Pop Quiz 5.1: Check your understanding!

Can you find a positive and a negative system example of each of the properties listed above? For example, for the memory property, the “positive” would be a system that has memory, and the “negative” would be a system that is memoryless.

*Solution on page 15*

A common type of system that we will end up studying is a causal stable LTI system. Causality is important because we will be studying time-domain signals. To ensure that signals don't blow up to  $\infty$ , we require stability. Finally, LTI systems, or linear time-invariant systems, will be the central class of systems we will study due to their nice properties and because many useful practical signals can be modeled as LTI systems.

## 5.1 Example: LTI system response

By using linearity and time-invariance, we can analyze the response of an LTI system to inputs and initial conditions. Consider the system with three different inputs:  $x_1(t)$ ,  $x_2(t)$ , and  $x_3(t)$ . Define the output of the system for each of the signals as  $y_1(t)$ ,  $y_2(t)$ , and  $y_3(t)$  respectively. Let us assume that the system is also affected by its initial conditions  $q_1(0)$  and  $q_2(0)$ .

We denote  $y_i^{\text{ZS}}(t)$  as the zero-state output due to  $x_i(t)$  with all initial conditions set to zero. Further, we denote  $y_{q_1}(t)$  and  $y_{q_2}(t)$  as the zero-input outputs produced by initial conditions in those two modes (with external inputs,  $x_i$ , set to zero).

Then, for any scalars  $\alpha_1, \dots, \alpha_5$ ,

$$x(t) = \alpha_1 x_1(t) + \alpha_2 x_2(t) + \alpha_3 x_3(t), \quad q(0) = \alpha_4 q_1(0) + \alpha_5 q_2(0)$$

$$\implies y(t) = \alpha_1 y_1^{\text{ZS}}(t) + \alpha_2 y_2^{\text{ZS}}(t) + \alpha_3 y_3^{\text{ZS}}(t) + \alpha_4 y_{q_1}(t) + \alpha_5 y_{q_2}(t).$$

Note that it is essential that  $y_i^{\text{ZS}}$  are computed with zero initial conditions otherwise the initial contributions would be double-counted. Likewise,  $y_{q_1}, y_{q_2}$  are computed with zero external input.

Further, if one of the inputs is shifted in time, say  $x_3(t - t_0)$ , then by the time-invariance property of the system, the output is also shifted by the same amount:

$$x(t) = \alpha_1 x_1(t) + \alpha_2 x_2(t) + \alpha_3 x_3(t - t_0), \quad q(0) = \alpha_4 q_1(0) + \alpha_5 q_2(0)$$

$$\implies y(t) = \alpha_1 y_1^{\text{ZS}}(t) + \alpha_2 y_2^{\text{ZS}}(t) + \alpha_3 y_3^{\text{ZS}}(t - t_0) + \alpha_4 y_{q_1}(t) + \alpha_5 y_{q_2}(t).$$

## 5.2 Example: LTI system properties to compute system response

Consider a continuous-time LTI system with input  $x(t)$  and output  $y(t)$ . We have the following information about the system:

$$\begin{aligned} x_1(t) = \cos(t) &\longrightarrow y_1(t) = \frac{1}{10} \cos(t) - \frac{3}{10} \sin(t), \\ x_2(t) = \cos(2t) &\longrightarrow y_2(t) = \frac{1}{5} \cos(2t) + \frac{3}{10} \sin(2t), \\ x_3(t) = \delta(t) &\longrightarrow y_3(t) = \delta(t), \end{aligned}$$

Then, for a new input

$$x_{\text{new}}(t) = \cos(2t) - 10 \sin(t + 10),$$

our goal is to find  $y_{\text{new}}(t)$  using only linearity and time invariance properties of the system.

Let us start by noting that since  $\sin(t) = \cos(t - \frac{\pi}{2})$ , we can use time-invariance of the system to write the output for  $\sin(t)$  (say  $y_4(t)$ ) as

$$y_4(t) = y_1\left(t - \frac{\pi}{2}\right) = \frac{1}{10} \sin(t) + \frac{3}{10} \cos(t).$$

Then, the output to  $\sin(t + 10)$  is (say  $y_5(t)$ ), again due to time-invariance is obtained by shifting  $y_4(t)$  by 10:

$$y_5(t) = \frac{1}{10} \sin(t + 10) + \frac{3}{10} \cos(t + 10)$$

Putting it all together using the linearity of the system, we get the desired output  $y_{\text{new}}(t)$ :

$$y_{\text{new}}(t) = \frac{1}{5} \cos(2t) + \frac{3}{10} \sin(2t) - \sin(t + 10) - 3 \cos(t + 10).$$

Finally, if the system has an initial condition at  $t = 0$ , superpose the corresponding measured responses:

$$y_{\text{new,IC}}(t) = y_{\text{new}}(t) + a \delta(t),$$

with  $a \in \mathbb{R}$  set by the magnitude of the initial condition.

## Pop Quiz Solutions

### Pop Quiz 2.1: Solution(s)

If you define the system as a multiplication of the input signal with  $\operatorname{Re}\{Ae^{j\omega_0 t}\}$ , you will see that the output signal has a frequency that is shifted by  $\omega_0$  from the input signal.

### Pop Quiz 2.2: Solution(s)

The 0.5 Hz modulation is closest to the original audio signal. The frequency modulation effect on audio can be described as a shift in the frequency spectrum of the original signal, resulting in a change in the perceived pitch and timbre of the sound, which ends up sounding like a reverberation.

### Pop Quiz 2.3: Solution(s)

See Problem 2 on Problem Set #3.

### Pop Quiz 4.1: Solution(s)

The two signals are the real and imaginary parts of the complex exponential:  $\operatorname{Re}\{x(t)\} = \cos(\omega t)$  and  $\operatorname{Im}\{x(t)\} = \sin(\omega t)$ .

### Pop Quiz 4.2: Solution(s)

Set  $\omega = 0$  and  $\sigma = 0$  to get  $x(t) = A = k$ . For real  $k$ , choose  $A \in \mathbb{R}$ .

### Pop Quiz 4.3: Solution(s)

Set  $\sigma > 0$ ,  $\omega = 0$ , and  $A \in \mathbb{R}$  to get  $x(t) = Ae^{\sigma t}$ , which grows unbounded as  $t \rightarrow \infty$ .

### Pop Quiz 4.4: Solution(s)

Set  $\sigma > 0$ ,  $\omega \neq 0$ , and  $A = |A|e^{j\phi}$  to get  $x(t) = |A|e^{\sigma t} \cos(\omega t + \phi)$ , which grows unbounded as  $t \rightarrow \infty$ .

### Pop Quiz 5.1: Solution(s)

Here are some examples:

- Linearity: Positive - Amplifier system  $y(t) = \alpha x(t)$ ; Negative - Clipping system  $y(t) = \text{clip}(x(t))$  is nonlinear.
- Time-invariance: Positive - Delay system  $y(t) = x(t - t_0)$ ; Negative - Time-varying gain system  $y(t) = tx(t)$ .
- Causality: Positive - Delay system  $y(t) = x(t - t_0)$ ; Negative - Anticipatory system  $y(t) = x(t + t_0)$  for  $t_0 > 0$ .
- Memory: Positive - Integrator system  $y(t) = \int_{-\infty}^t x(\tau) d\tau$ ; Negative - Memoryless system  $y(t) = x(t)^2$  only depends on the current instant of time.
- Invertibility: Positive - Scaling system  $y(t) = 2x(t)$ ; Negative - Clipping system  $y(t) = \text{clip}(x(t))$  cannot be inverted as all values lead to the same clipped behavior after a certain threshold.
- Stability: Positive - An RC circuit with decaying voltage:  $y(t) = (1 - e^{-t/\tau})x(t)$ ; Negative - Unstable system (e.g.,  $y(t) = e^t x(t)$ ).